Python Programming

Beginner Study Notes and Practice Guide

This short guide introduces core concepts and provides practice ideas. It is an original learning summary, not a substitute for a complete textbook.

Introduction

Python is a general-purpose language known for readable syntax and a broad ecosystem.

Variables and types

Variables refer to values. Common types include int, float, str, bool, list, tuple, and dict.

Input and output

Use input() to receive text and print() to display a result. Convert input with int() or float() when numeric values are needed.

Conditions and loops

Use if/elif/else for decisions, and for or while loops for repetition.

Functions

Define a function with def. Parameters provide input, and return sends a result back to the caller.

Collections

Lists are ordered and mutable; tuples are ordered and immutable; dictionaries store key-value pairs.

Practice

Write a program that counts vowels in a string and another that stores book titles in a list.

Practice checklist

- Review the concepts and write a small example.
- Run your example and test an edge case.
- Explain the solution in your own words.